

SS Prerequisite: Construction Activity Pollution Prevention

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core and Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses and Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To reduce pollution from construction activities by controlling soil erosion, waterway sedimentation, and airborne dust that disproportionately impact frontline communities.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Create and implement an erosion and sedimentation control plan for all construction activities associated with the project. The plan must conform to the erosion and sedimentation requirements of the 2017 U.S. Environmental Protection Agency (EPA) Construction General Permit (CGP) or local equivalent, whichever is more stringent. Projects must apply the CGP regardless of size. The plan must describe the measures implemented.

GUIDANCE

There are no substantive changes to the prerequisite requirements; refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Updated the referenced standard which was out of date.

Step-by-Step Guidance

- ▶ Refer to the 2017 edition of the U.S. Environmental Protection Agency (EPA) Construction General Permit (CGP) where any instance of the outdated standard is mentioned.

Further Explanation

Referenced Standards

- ▶ U.S. EPA, National Pollutant Discharge Elimination System (NPDES), Stormwater Discharges from Construction Activities, 2017: <https://www.epa.gov/npdes/stormwater-discharges-construction-activities>

Connection to Ongoing Performance:

- ▶ LEED O+M SS credit Rainwater Management: Implementing an ESC plan that minimizes soil compaction where vegetation will be planted or where infiltration measures will be installed will support reducing runoff volumes, in accordance with the related credit's requirements.

SS Prerequisite: Environmental Site Assessment

This prerequisite applies to

- ▶ BD+C: Schools
- ▶ BD+C: Healthcare

INTENT

To protect the health of vulnerable populations by ensuring that the site is assessed for environmental contamination and that any environmental contamination has been remediated.

REQUIREMENTS

SCHOOLS, HEALTHCARE

Conduct a Phase I Environmental Site Assessment as described in ASTM E1527-13 (or a local equivalent) to determine whether environmental contamination exists at the site. If contamination is suspected, conduct a Phase II Environmental Site Assessment as described in ASTM E1903-11 (or a local equivalent).

If a site is contaminated, remediate the site to meet local, state, or national environmental protection agency region residential (unrestricted) standards, whichever are most stringent.

GUIDANCE

There are no substantive changes to the prerequisite requirements; refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Updated the referenced standard which was out of date.

Step-by-Step Guidance

- ▶ Refer to the E1527-13 edition of the ASTM standard where any instance of the outdated standard is mentioned.

Further Explanation

Referenced Standards

- ▶ ASTM E1527-13 Standard Practice for Environmental Site Assessments: Phase I Environmental Site Assessment Process

SS Credit: Site Assessment

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core and Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses and Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To assess site conditions, environmental justice concerns, and cultural and social factors, before design to evaluate sustainable options and inform related decisions about site design.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Complete and document a site survey or assessment² that includes the following information:

Topography.

- ▶ Contour mapping
- ▶ Unique topographic features
- ▶ Slope stability risks

Hydrology.

- ▶ Special Flood Hazard Areas (SPFHA) as determined by FEMA's Flood Insurance Rate Map (FIRM) (or local equivalent for projects outside the U.S.)
- ▶ Delineated natural water bodies wetlands, lakes, streams, and shorelines (refer to U.S. EPA's Clean Water Act or local equivalent for projects outside the U.S.)
- ▶ Rainwater collection and reuse opportunities
- ▶ Impervious and pervious surfaces within the site boundary

Climate.

- ▶ Solar exposure and shading opportunities
- ▶ Heat island effect potential
- ▶ Seasonal sun angles
- ▶ Prevailing winds
- ▶ Average monthly precipitation and temperature ranges

Vegetation.

- ▶ Primary vegetation types
- ▶ Greenfield area
- ▶ Significant tree mapping
- ▶ Federal or state threatened or endangered species lists; for projects outside the U.S., International Union for Conservation of Nature (IUCN) Red List of Threatened Species
- ▶ Invasive plant species listed by regional, state, or federal entities
- ▶ EPA Level III ecoregion description (or local equivalent)

² Components adapted from the Sustainable Sites Initiative: Guidelines and Performance Benchmarks 2009, Prerequisite 2.1: Site Assessment.

Soils.

- ▶ Natural Resources Conservation Service soils delineation (or local equivalent for projects outside the U.S.)
- ▶ U.S. Natural Resources Conservation Service (or local equivalent for projects outside the United States) prime farmland, unique farmland, farmland of statewide importance, or farmland of local importance
- ▶ Healthy soils
- ▶ Previous development
- ▶ Disturbed soils

Human use.

- ▶ Views
- ▶ Adjacent transportation infrastructure, bicycle network, and bicycle storage
- ▶ Adjacent diverse uses
- ▶ Construction materials with existing recycle or reuse potential

Human health effects.

- ▶ Proximity of vulnerable populations
- ▶ Adjacent physical activity opportunities
- ▶ Proximity to major sources of air and water pollution

The survey or assessment should demonstrate the relationships between the site features and topics listed above and how these features influenced the project design; give the reasons for not addressing any of those topics.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The removal of a U.S.-specific standard, TR-55, from the required considerations and broadening of this concept to account for all impervious and pervious surfaces on site has made this updated credit more approachable to most project teams. The credit now requires projects to describe their EPA Level III Ecoregion (or local equivalent) in order to understand what native and adapted vegetation is appropriate for their site, integrative information that better contextualizes the credit intent.

Further Explanation

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Hydrology: Disregard any mention of the Natural Resources Conservation Service TR-55 program as it is no longer required.
- ▶ Hydrology: Estimate the water storage capacity of the site by calculating the area of impervious and pervious surfaces. Table 2 lists of other possible sources of information.
- ▶ Vegetation:
 - Source – EPA Level III Ecoregion Descriptions (ftp://newftp.epa.gov/EPADDataCommons/ORD/Ecoregions/pubs/NA_TerrestrialEcoregionsLevel3_Final-2june11_CEC.pdf)
 - Description – Descriptions identifying North American ecoregions and detailing their associated ecosystems and vegetation types.

Required Documentation

- ▶ Provide a map illustrating the topography of the site.
- ▶ Provide a map illustrating the site's Special Flood Hazard Areas (SPFHA) as determined by FEMA's Flood Insurance Rate Map (FIRM) (or local equivalent showing the 100-year floodplain for projects outside the U.S.).
- ▶ Provide the description of the site's EPA Level III ecoregion (or local equivalent).

Referenced Standards

- ▶ EPA Level III Ecoregion Descriptions:
ftp://newftp.epa.gov/EPADDataCommons/ORD/Ecoregions/pubs/NA_TerrestrialEcoregionsLevel3_Final-2june11_CEC.pdf

Connection to Ongoing Performance:

- ▶ LEED O+M SS credit Site Management: Conducting a site assessment and identifying natural areas providing habitat will fulfill the related credit's site assessment requirements.
- ▶ LEED O+M LT credit Transportation Performance: Analyzing the surrounding sites and diverse uses, transportation infrastructure, bicycle network, as well as assessing existing bicycle facilities and potential future facility needs, may help inform and influence the improvement of a project's transportation performance score.
- ▶ LEED O+M SS credit Rainwater Management: Studying the climate, rainfall, and hydrology of the site and watershed will help determine applicable strategies to earn the related performance-based credit.
- ▶ LEED O+M SS credit Heat Island Reduction: Site assessment can lead to identification of paving, shading, or roofing materials that can contribute to requirements of the related performance-based credit.
- ▶ LEED O+M EA credit Energy Performance: An analysis of the climate, including solar access, temperatures, diurnal swings, wind patterns, humidity, and rainfall will support more effective passive and active energy efficiency strategies.

SS Credit: Protect or Restore Habitat

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core and Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses and Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1 point)

INTENT

To conserve existing natural areas and restore damaged areas to provide habitat and promote biodiversity.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Preserve and protect from all development and construction activity 40% of the *greenfield* area on the site (if such areas exist).

AND

Restore a portion of the site, including the building footprint, identified as *previously disturbed (if such areas exist)* and follow vegetation and soil requirements below. Vegetated roof surfaces may be included in the habitat area calculations if the plants are native or adapted and provide habitat. Points are awarded according to Table 1.

Table 1. Points for percentage of area restored

Restored Area	Points BD+C (except Healthcare)	Points BD+C (Healthcare)
15% of previously disturbed area	1	1
25% of previously disturbed area	2	-

Soil Restoration

Restore disturbed soils in areas that will later serve as the final habitat area.

- ▶ Imported soils may not include the following:
 - soils defined regionally by the Natural Resources Conservation Service web soil survey (or local equivalent for projects outside the U.S.) as *prime farmland*, unique farmland, or farmland of statewide or local importance;
 - soils from other *greenfield* sites; or
 - sphagnum peat moss

Vegetation

Plant a minimum of 6 species of vegetation that are native or adapted to the project's EPA Level III ecoregion (or local equivalent for projects outside of the U.S.). Include a minimum of 2 out of the following plant categories: tree, shrub, and ground cover. Designate a portion of the habitat area for a pollinator garden consisting of native flowering plants and totaling at least 30 square feet (3 square meters).

Schools only:

Dedicated athletic fields that are solely for athletic uses are exempted from counting toward the total site area. These areas may not count toward the protected greenfield or restored habitat areas.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

This updated credit is now more achievable for various project types and more directly addresses the intent of the credit as it relates to habitat protection and revitalization. The restoration requirement was reduced from a minimum of 30% to 15%, and a second threshold was added at 25% for two points to reward projects for increased restoration. Soil requirements have been streamlined and testing requirements have been removed. A vegetation section was added to encourage planting of diverse vegetation and pollinator habitat. Option 2. Financial Support was removed to encourage focus on local habitat restoration.

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following modifications:

Verify that at least 15%, rather than 30%, of previously disturbed areas will be restored.

- ▶ Soil tests are no longer required.
- ▶ Option 2 Financial Support is no longer offered in this credit.
- ▶ Vegetation:
 - A contiguous series of habitat areas, including pollinator gardens, connected by landscaped paths are encouraged in order to facilitate connectivity for wildlife, especially pollinating insects that cannot travel far without access to food sources. Pollinator gardens are recommended for areas such as parking lots, which include several small contiguous parcels of vegetation.
 - *Ground cover* includes native and adapted grasses as well as perennials and other plant species. See WE Credit Outdoor Water Use Reduction for more guidance on ground cover.
 - Refer to the Xerces Society's Pollinator-Friendly Plant Lists as a resource to select appropriate plants for local pollinators.

Further Explanation

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Best Practices for Habitat Protection:
 - For more information on invasive vegetation, refer to the USDA's NRCS PLANTS Database, the Invasive Plant Atlas of the United States website, and/or state and federal Noxious Weed laws (or local equivalent for projects outside of the U.S.).
- ▶ Vegetated Roofs:
 - Projects no longer need to meet a floor-area ratio (FAR) density minimum for vegetated roofs to be considered part of the restored area.

Required Documentation

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Site plan depicting the project boundary, building footprint(s), preserved greenfield area(s) (if applicable), previously disturbed area, restored habitat area, native and adapted vegetation, and designated pollinator garden.

- ▶ Plants list, including pollinator plants, appropriate for project's EPA Level III ecoregion (or local equivalent for projects outside the U.S.); must include a minimum of 2 out of the following plant categories: tree, shrub, and ground cover.
- ▶ Description of disturbed or compacted soils to be revegetated
- ▶ EPA Level III Ecoregion description (or local equivalent)
- ▶ No longer required: Soil test results.
- ▶ No longer required: Projects with vegetated roofs to provide the floor area ratio (FAR)
- ▶ Option 2: Financial Support has been removed and relevant documentation requirements are no longer applicable.

Referenced Standards

Refer to the LEED v4 reference guide, with the following additions:

- ▶ **EPA Level III Ecoregion Descriptions:**
ftp://newftp.epa.gov/EPADDataCommons/ORD/Ecoregions/pubs/NA_TerrestrialEcoregionsLevel3_Final-2june11_CEC.pdf
- ▶ **USDA Natural Resources Conservation Service, PLANTS Database:**
<https://www.nrcs.usda.gov/wps/portal/nrcs/main/national/plantsanimals/plants/>
- ▶ **Invasive Plant Atlas of the United States:** <https://www.invasiveplantatlas.org/>
- ▶ **U.S. State and Federal Noxious Weed laws:** <https://plants.usda.gov/java/noxComposite>
- ▶ **Xerces Society for Invertebrate Conservation, Pollinator-Friendly Plant Lists:**
xerces.org/pollinator-conservation/plant-lists/

Exemplary Performance

- ▶ Restore a minimum of 35% (including the building footprint) of all previously disturbed area on site using soil and vegetation that meet the credit requirements.

Definitions

- ▶ **adapted vegetation** vegetation that is not native to the particular region it was introduced to, but has evolved or maintained characteristics conducive for healthy growth and requires no additional resources or maintenance, such as water for irrigation, in comparison to similar species native to the area. An adapted species is non-aggressive or disruptive to native plant communities.
- ▶ **contiguous** adjoined or adjacent parcels that facilitate connectivity
- ▶ **greenfield site** area that has not been graded, compacted, cleared, or disturbed and that supports (or could support) open space, habitat, or natural hydrology. See also: previously disturbed.
- ▶ **habitat area** area that supports native wildlife populations by providing locations for nesting, cover, respite, and/or food sources.
- ▶ **invasive vegetation** nonnative or exotic vegetation that quickly grows and aggressively reproduces, often disrupting or displacing native plant communities. The plant's vigor combined with a lack of natural enemies leads to outbreak populations.
- ▶ **pollinator** an insect or animal that moves pollen from the male anther of a flower to the female stigma of a flower in order to pollinate the vegetation.
- ▶ **pollinator garden** area that is planted with native flowering vegetation that provide nectar, pollen, habitat, or nesting area for native pollinating fauna.
- ▶ **previously disturbed** areas that have been graded, compacted, cleared, previously developed, or disturbed in any way. These are areas that do not qualify as 'greenfield.'

Connection to Ongoing Performance:

- ▶ LEED O+M credit Site Management: Naturally vegetated areas that meet the requirements of this credit are less likely to require the routine use of maintenance equipment. They could also reduce irrigation, nutrient application, and erosion, thereby helping the project meet the requirements of the related performance-based credit.
- ▶ LEED O+M SS credit Rainwater Management: Restoring and protecting greenfield areas with native or adapted vegetation will help provide applicable strategies to earn the related performance-based credit.
- ▶ LEED O+M SS credit Heat Island Reduction: Vegetated roofs can be counted for both credits.

SS Credit: Open Space

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core and Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses and Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To create exterior open space that encourages interaction with the environment, social interaction, passive recreation, and physical activities.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Provide outdoor space greater than or equal to 30% of the total site area (including building footprint).

At least 25% of the required outdoor open space must be vegetated space planted with two or more types of vegetation or have *overhead vegetated canopy*.

The outdoor space must be physically accessible and be one or more of the following:

- ▶ social area: a pedestrian-oriented paving or landscape area that accommodate outdoor social activities;
- ▶ recreational area: a recreation-oriented paving or landscape area that encourage physical activity;
- ▶ diverse green space: a landscape area with two or more types of vegetation that provide opportunities for year-round visual interest;
- ▶ garden: a garden space dedicated to community gardens or urban food production; or
- ▶ habitat area: preserved or created habitat that meets the criteria of SS Credit Protect or Restore Habitat and includes elements of human interaction. These areas automatically meet the vegetation criteria of this credit.

Extensive or intensive vegetated roofs that are physically accessible can be used toward the minimum vegetation requirement, and qualifying roof-based physically accessible paving areas can be used toward credit compliance.

Wetlands or naturally designed ponds may count as open space if the side slope gradients average 1:4 (vertical:horizontal) or less and are vegetated.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions:

Behind the Intent

Beta Update

Updates clarify requirements about the percentage of outdoor space and vegetated space to qualify as open space. Projects must provide outdoor space greater than or equal to 30% of the total site area. Of the total outdoor open space required, a minimum of 25% must be vegetated or have overhead vegetated canopy. Minimal changes to verbiage regarding vegetation have also been made to clarify types of vegetation that meet the requirements.

Further Explanation

Connection to Ongoing Performance:

- ▶ LEED O+M SS credit Rainwater Management: Vegetated landscape areas with native or adapted vegetation will help provide applicable strategies to earn the related credit.
- ▶ LEED O+M SS credit Heat Island Reduction: Vegetated roofs can be counted for both credits.

SS Credit: Rainwater Management

This credit applies to

- ▶ BD+C: New Construction (1-3 points)
- ▶ BD+C: Core and Shell (1-3 points)
- ▶ BD+C: Schools (1-3 points)
- ▶ BD+C: Retail (1-3 points)
- ▶ BD+C: Data Centers (1-3 points)
- ▶ BD+C: Warehouses and Distribution Centers (1-3 points)
- ▶ BD+C: Hospitality (1-3 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To reduce runoff volume and improve water quality by replicating the natural hydrology and water balance of the site, based on historical conditions and undeveloped ecosystems in the region to avoid contributing to flooding downstream in frontline communities.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Option 1. Percentile of Rainfall Events (1-3 points except Healthcare, 1-2 points Healthcare)

In a manner best replicating *natural site hydrology* processes, retain *on site* the runoff from the associated percentile of regional or local rainfall events. The percentile event volume must be retained (i.e. infiltrated, evapotranspired, or collected and reused) using *low-impact development (LID)* and *green infrastructure (GI) practices*. GI and LID strategies can be either structural or non-structural. Points are awarded according to Table 1.

Table 1. Points for percentile of rainfall retained

All Projects	Zero lot line Projects	Points	Points Healthcare
80 th Percentile	70 th Percentile	1	1
85 th Percentile	75 th Percentile	2	2
90 th Percentile	80 th Percentile	3	-

OR

Option 2. Natural land cover conditions (3 points except Healthcare, 2 points Healthcare)

Calculate the difference between the projected runoff volume under the proposed design conditions and the runoff volume under natural land cover conditions that existed prior to any disturbance. Retain (i.e. infiltrate, evapotranspire, or collect and reuse) on site the increase in runoff volume using LID and GI *practices*.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Projects are now able to earn points for retaining runoff for the 80th, 85th, or 90th percentile of regional or local rainfall events, whereas the credit previously required the 95th or 98th percentile.

Thresholds for zero lot line projects have also been reduced. To ensure clarity, a more refined definition for zero lot line has been included and the term “manage” has been replaced with the more specific and technical “retain (i.e. infiltrate, evapotranspire, or collect and reuse)”. More guidance and resources have also been added to clarify acceptable LID strategies. New exemplary performance thresholds for all projects have been added, including exemplary performance under Option 2. Natural landcover conditions which rewards projects for fostering resilience by designing for future precipitation conditions.

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ The percentile rainfall event thresholds have been changed to 80th, 85th, and 90th percentile for all projects and 70th, 75th, and 80th percentile for zero lot line projects.
- ▶ Option 1, Step 2. Obtain Rainfall Data for Project Location: Use daily rainfall data and the methodology in the U.S. Environmental Protection Agency (EPA) Technical Guidance on Implementing the Stormwater Runoff Requirements for Federal Projects under Section 438 of the Energy Independence and Security Act to determine the percentile amount to be retained.
- ▶ Option 1, Step 2. Obtain Rainfall Data for Project Location: Once you’ve gathered the rainfall events data, any rainfall events less than or equal to 0.1 inches should be omitted.
- ▶ Option 1, Step 7. Conventional grey infrastructure devices and hard runoff conveyance systems, such as detention or retention ponds, pipes, culverts and vaults, can be compliant if used as a component of LID/GI systems to meet the requirements of this credit.

Further Explanation

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Examples of acceptable LID techniques include the following:
 - planting rain gardens with native or adapted plant material (e.g. trees or shrubs);
 - installing a vegetated roof;
 - using permeable paving, consisting of porous above-ground materials (e.g., open pavers, engineered products), a base layer designed to drain water away from the building, and (often) a 6-inch-deep (150 millimeters) subbase; and
 - installing permanent infiltration or collection features (e.g., vegetated swale, rain garden, rainwater cistern) that can retain the runoff from the associated percentile of regional or local rainfall events.
- ▶ Treatment of Runoff:

Projects teams should consider prioritizing infiltration of runoff from impervious surfaces considered to be a significant source of pollutants in rainwater runoff. Such surfaces include those which are subject to: vehicular use; industrial activities; or storage of erodible or leachable materials, wastes, or chemicals and which receive direct rainfall or the run-on of rainfall.
- ▶ Green Infrastructure and Low-Impact Development Strategies:

Employ LID and GI strategies that mimic the natural infiltration-based, groundwater-driven hydrology of the site. If project teams are facing environmental or technical barriers to proceeding with LID techniques, consult the EPA’s Low-Impact Development website for detailed “Barrier Buster Fact Sheets” that address specific issues, such as how to combat clay soil constraints and designing LID practices on sloped sites.
- ▶ Project Type Variations, zero lot line projects:

Any setback area resulting from a planned below-grade space, such as an underground parking garage, does not void a project’s zero lot line status so long as the setback area is designed as pedestrian-oriented amenity space, such as outdoor seating.

Referenced Standards

- ▶ EPA’s LID website: <https://www.epa.gov/nps/lid>

- ▶ **Georgetown Climate Center's Green Infrastructure Toolkit:**
<http://www.georgetownclimate.org/adaptation/toolkits/green-infrastructure-toolkit/green-infrastructure-strategies-and-techniques.html>

Exemplary Performance

Option 1. Percentile of rainfall events

All Projects	Zero lot line Projects
98 th Percentile	95 th Percentile

Option 2. Natural land cover conditions

Retain on site an additional 20% in runoff volume to address future precipitation rates (1.2 * [difference in annual runoff volume from the natural landcover to the post-developed condition]) using LID and GI practices.

Definitions

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ **retain (rainwater) on site** to capture and reserve a specified volume of rainfall to mimic natural hydrologic function. Retention is a function of rainwater management that includes strategies involving evapotranspiration, infiltration, and capture and reuse.

Connection to Ongoing Performance:

- ▶ LEED O+M WE credit Water Performance: LID and GI measures that harvest and reuse rainwater may help reduce potable water demand. Thus, these measures help ensure that projects will use less water throughout the building life cycle, which may help improve a project's water performance score. Tracking water consumption on a regular basis through metering supports effective water management and provides performance data to help verify that building systems are operating as designed.
- ▶ LEED O+M SS credit Heat Island Reduction: Vegetated roofs installed for achievement of this credit will also qualify for the related performance-based credit.

SS Credit: Heat Island Reduction

This credit applies to

- ▶ BD+C: New Construction (1-4 points)
- ▶ BD+C: Core and Shell (1-4 points)
- ▶ BD+C: Schools (1-4 points)
- ▶ BD+C: Retail (1-4 points)
- ▶ BD+C: Data Centers (1-4 points)
- ▶ BD+C: Warehouses and Distribution Centers (1-4 points)
- ▶ BD+C: Hospitality (1-4 points)
- ▶ BD+C: Healthcare (1-4 points)

INTENT

To minimize inequitable effects on microclimates and human, especially frontline communities, and wildlife habitats by reducing heat islands.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Choose one of the following options:

Option 1. Nonroof and Roof (2 points except Healthcare, 1 point Healthcare)

Meet the following criterion:

$$\begin{array}{ccccccc} \text{Area of Nonroof Measures} & & \text{Area of High-Reflectance Roof} & & \text{Area of Vegetated Roof} & & \\ \text{-----} & + & \text{-----} & + & \text{-----} & \geq & \text{Total Site Paving Area} + \text{Total Roof Area} \\ 0.5 & & 0.75 & & 0.75 & & \end{array}$$

Alternatively, an SRI and SR weighted average approach may be used to calculate compliance.

Use any combination of the following strategies.

Nonroof Measures

- ▶ Use the existing plant material or install plants that provide shade over paving areas (including playgrounds) on the site within 10 years of planting. Install vegetated planters. Plants must be in place at the time of occupancy permit and cannot include artificial turf.
- ▶ Provide shade with structures covered by energy generation systems, such as solar thermal collectors, photovoltaics, and wind turbines.
- ▶ Provide shade with architectural devices or structures. If the device or structure is a roof, it shall have an aged *solar reflectance (SR)* value of at least 0.28 as measured in accordance with ANSI/CRRC S100. If the device or structure is not a roof, or if aged solar reflectance information is not available, it shall have at installation an initial SR of at least 0.33 as measured in accordance with ANSI/CRRC S100.
- ▶ Provide shade with vegetated structures.
- ▶ Use paving materials with an initial *solar reflectance (SR)* value of at least 0.33.
- ▶ Use an *open-grid pavement system* (at least 50% unbound).

High-Reflectance Roof

Use roofing materials that have an aged SRI equal to or greater than the values in Table 1. If aged SRI is not available, the roofing material shall have an initial SRI equal to or greater than the values in Table 1.

Table 1. Minimum solar reflectance index value, by roof slope

	Slope	Initial SRI	Aged SRI
Low-sloped roof	≤ 2:12	82	64
Steep-sloped roof	> 2:12	39	32

Roof area that consists of functional, usable spaces (such as helipads, recreation courts, and similar amenity areas) may meet the requirements of nonroof measures. Applicable roof area excludes roof area covered by mechanical equipment, solar energy panels, skylights, and any other appurtenances.

Vegetated Roof

Install a vegetated roof using native or adapted plant species.

OR

Option 2. Parking under Cover (1 point)

Place a minimum of 75% of *parking spaces under cover*. Any roof used to shade or cover parking must (1) have an aged SRI of at least 32 (if aged value information is not available, use materials with an initial SRI of at least 39 at installation), (2) be a vegetated roof, or (3) be covered by energy generation systems, such as solar thermal collectors, photovoltaics, and wind turbines.

The credit calculations must include all existing and new off-street parking spaces that are leased or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

This updated credit references the new ANSI/Cool Roof Rating Council S100 standard for “Rapid Ratings.” Updates also clarify that functional space for occupant use located on a roof can be included under nonroof calculations. To ensure that this credit is maximizing its intended impact, projects pursuing Option 2. Parking Under Cover are required to include all off-street parking used by the project in their calculations.

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following modification:

- ▶ Identify hardscape and roof area
 - Common roads that serve multiple buildings should not be included in this calculation. If the roads include the primary parking areas (i.e. parking spots along the road), those parking areas should be included in the calculation.
- ▶ Selecting roofing materials:
 - Initial SRI shall be computed in accordance with ASTM E1980, under medium wind-speed conditions from initial values of solar reflectance and thermal emittance measured in accordance with ANSI/CRRC S100.
 - Aged SRI shall be computed in accordance with ASTM E1980, under medium wind-speed conditions from aged values of solar reflectance and thermal emittance measured in accordance with ANSI/CRRC S100.
 - Values of roof solar reflectance, thermal emittance, and solar reflectance index may be available at the Cool Roof Ratings Council's Rated Products Directory.
- ▶ Evaluate compliance against credit requirements:

- Equation 2, within the “summed for all high-reflectance roof areas” portion, replace ‘Required SR’ in the denominator with ‘Required SRI’.
- Credit calculations must include all existing and new off-street parking spaces owned or leased by the project.

Further Explanation

Referenced Standards

- ▶ ANSI/CRRRC S100, Standard Test Methods for Determining Radiative Properties of Material
- ▶ ASTM E1980 Standard Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces
- ▶ Cool Roof Ratings Council, Rated Products Directory: <https://coolroofs.org/directory>

Definitions

- ▶ **common road** for the purposes of this credit, a common road is defined as a narrow or small road or alleyway between buildings, which may or may not be drivable.
- ▶ **solar reflectance (SR)** The ratio of the reflected solar flux to the incident solar flux.
- ▶ **solar reflectance, initial (initial SR)** a solar reflectance that is measured prior to aging.
- ▶ **solar reflectance, aged (aged SR)** a solar reflectance that is measured after laboratory or natural aging.
- ▶ **solar reflectance index (SRI)** a measure of the constructed surface’s ability to stay cool in the sun by reflecting solar radiation and emitting thermal radiation. A standard black surface has an initial SRI of 0, and a standard white surface has an initial SRI of 100.
- ▶ **thermal emittance (TE)** the ratio of the radiant heat flux emitted by a specimen to that emitted by a blackbody radiator at the same temperature.
- ▶ **thermal emittance, initial (initial TE)** a thermal emittance that is measured prior to aging.
- ▶ **thermal emittance, aged (aged TE)** a thermal emittance that is measured after laboratory or natural aging.

Connection to Ongoing Performance:

- ▶ LEED O+M SS Credit Rainwater Management: Reducing a site’s hardscape area and/or using open-grid paving will improve infiltration rates and may help projects earn the related performance-based credit. Vegetated roofs can also contribute to rainwater management.

SS Credit: Light Pollution Reduction

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core and Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses and Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To increase night sky access, improve nighttime visibility, and reduce the consequences of development for wildlife and people.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Meet uplight and light trespass requirements, using either the backlight-uplight-glare (BUG) method (Option 1) or the calculation method (Option 2). Projects may use different options for uplight and light trespass.

Meet these requirements for all exterior luminaires located inside the project boundary (except those listed under “Exemptions”), based on the following:

- ▶ the photometric characteristics of each luminaire when mounted in the same orientation and tilt as specified in the project design; and
- ▶ the lighting zone of the project property (at the time construction begins). Classify the project under one lighting zone using the lighting zones definitions provided in the Illuminating Engineering Society and International Dark Sky Association (IES/IDA) Model Lighting Ordinance (MLO) User Guide.

Additionally, meet the internally illuminated signage requirement.

Uplight

Option 1. BUG Rating Method

Do not exceed the following luminaire uplight ratings, based on the specific light source installed in the luminaire, as defined in IES TM-15-11, Addendum A.

Table 1. Maximum uplight ratings for luminaires

MLO lighting zone	Luminaire uplight rating
LZ0	U0
LZ1	U1
LZ2	U2
LZ3	U3
LZ4	U4

OR

Option 2. Calculation Method

Do not exceed the following percentages of total lumens emitted above horizontal.

Table 2. Maximum percentage of total lumens emitted above horizontal, by lighting zone

MLO lighting zone	Maximum allowed percentage of total luminaire lumens emitted above horizontal
LZ0	0%
LZ1	0%
LZ2	1.5%
LZ3	3%
LZ4	6%

AND

Light Trespass

Option 1. BUG Rating Method

Do not exceed the following luminaire backlight and glare ratings (based on the specific light source installed in the luminaire), as defined in IES TM-15-11, Addendum A, based on the mounting location and distance from the lighting boundary.

Table 3. Maximum backlight and glare ratings

	MLO lighting zone				
Luminaire mounting	LZ0	LZ1	LZ2	LZ3	LZ4
	Allowed backlight ratings				
> 2 mounting heights from lighting boundary	B1	B3	B4	B5	B5
1 to 2 mounting heights from lighting boundary and properly oriented	B1	B2	B3	B4	B4
0.5 to 1 mounting height to lighting boundary and properly oriented	B0	B1	B2	B3	B3
< 0.5 mounting height to lighting boundary and properly oriented	B0	B0	B0	B1	B2
	Allowed glare ratings				
Building-mounted > 2 mounting heights from any lighting boundary	G0	G1	G2	G3	G4
Building-mounted 1-2 mounting heights from any lighting boundary	G0	G0	G1	G1	G2
Building-mounted 0.5 to 1 mounting heights from any lighting boundary	G0	G0	G0	G1	G1

Building-mounted < 0.5 mounting heights from any lighting boundary	G0	G0	G0	G0	G1
All other luminaires	G0	G1	G2	G3	G4

The lighting boundary is located at the property lines of the property, or properties, that the LEED project occupies. The lighting boundary can be modified under the following conditions:

- ▶ When the property line abuts a public area that includes, but is not limited to, a walkway, bikeway, plaza, or parking lot, the lighting boundary may be moved to 5 feet (1.5 meters) beyond the property line.
- ▶ When the property line abuts a public street, alley, or transit corridor, the lighting boundary may be moved to the center line of that street, alley, or corridor.
- ▶ When there are additional properties owned by the same entity that are contiguous to the property, or properties, that the LEED project is within and have the same or higher MLO lighting zone designation as the LEED project, the lighting boundary may be expanded to include those properties.

Orient all luminaires less than two mounting heights from the lighting boundary such that the backlight points toward the nearest lighting boundary line. Building-mounted luminaires with the backlight oriented toward the building are exempt from the backlight rating requirement.

OR

Option 2. Calculation Method

Do not exceed the following vertical illuminances at the lighting boundary (use the definition of lighting boundary in Option 1). Calculation points may be no more than 5 feet (1.5 meters) apart. Vertical illuminances must be calculated on vertical planes running parallel to the lighting boundary, with the normal to each plane oriented toward the property and perpendicular to the lighting boundary, extending from grade level to 33 feet (10 meters) above the height of the highest luminaire.

Table 4. Maximum vertical illuminance at lighting boundary, by lighting zone

MLO lighting zone	Vertical illuminance
LZ0	0.05 fc (0.5 lux)
LZ1	0.05 fc (0.5 lux)
LZ2	0.10 fc (1 lux)
LZ3	0.20 fc (2 lux)
LZ4	0.60 fc (6 lux)

FC = footcandle.

AND

Internally Illuminated Exterior Signage

Do not exceed a luminance of 200 cd/m² (nits) during nighttime hours and 2000 cd/m² (nits) during daytime hours.

Exemptions from Uplight and Light Trespass Requirements

The following exterior lighting is exempt from the requirements, provided it is controlled separately from the nonexempt lighting:

- ▶ specialized signal, directional, and marker lighting for transportation;
- ▶ lighting that is used solely for façade and landscape lighting in MLO lighting zones 3 and 4, and is automatically turned off from midnight until 6 a.m.;
- ▶ lighting for theatrical purposes for stage, film, and video performances;
- ▶ government-mandated roadway lighting;

- ▶ hospital emergency departments, including associated helipads;
- ▶ lighting for the national flag in MLO lighting zones 2, 3, or 4; and
- ▶ internally illuminated signage.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following modification:

- ▶ Existing Fixtures
 - Calculations must include existing fixtures. If fixture type or manufacturer cannot be identified, the project team should analyze a new, similar fixture with the same bulb type, wattage, lumen output, and physical design that can be expected to have similar photometric properties.

Further Explanation

Connection to Ongoing Performance:

- ▶ LEED O+M EA credit Energy Performance: A proper lighting design both minimizes light pollution and maximizes energy efficiency, which could improve a project's overall energy performance.

SS Credit: Site Master Plan

This credit applies to

- ▶ BD+C: Schools (1-4 points)

INTENT

To ensure that the sustainable site benefits achieved by the project continue, regardless of future changes in programs or demographics.

REQUIREMENTS

SCHOOLS

The project must achieve at least four of the following six credits, using the associated calculation methods. The achieved credits must then be recalculated using the data from the master plan.

- ▶ LT Credit: High Priority Site
- ▶ SS Credit: Site Development—Protect or Restore Habitat
- ▶ SS Credit: Open Space
- ▶ SS Credit: Rainwater Management
- ▶ SS Credit: Heat Island Reduction
- ▶ SS Credit: Light Pollution Reduction

A *site master plan* for the school must be developed in collaboration with school authorities. Previous sustainable site design measures should be considered in all master-planning efforts so that existing infrastructure is retained whenever possible. The master plan must therefore include current construction activity plus future construction (within the building's lifespan) that affects the site. The master plan development footprint must also include parking, paving, and utilities.

Projects where no future development is planned are not eligible for this credit.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance:

- ▶ LEED O+M SS credit Site Management: Considering how infrastructure included in the scope of the project can be maintained or enhanced will help with establishing best practice maintenance procedures. Upholding sustainable site design measures informed by the site master plan can potentially reduce the need for power maintenance equipment, facilitating achievement of the related performance-based credit.

SS Credit: Tenant Design and Construction Guidelines

This credit applies to

- ▶ BD+C: Core and Shell (1-4 points)

INTENT

To educate tenants in implementing sustainable design and construction features in their tenant improvement build-outs.

REQUIREMENTS

CS

Publish for tenants an illustrated document with the following content, as applicable:

- ▶ a description of the sustainable design and construction features incorporated in the core and shell project and the project's sustainability goals and objectives, including those for tenant spaces;
- ▶ recommendations, including examples, for sustainable strategies, products, materials, and services; and
- ▶ information that enables a tenant to coordinate space design and construction with the building systems when pursuing the following LEED v4.1 for Interior Design and Construction prerequisites and credits:
 - WE Prerequisite: Indoor Water Use Reduction
 - WE Credit: Indoor Water Use Reduction
 - EA Prerequisite: Minimum Energy Performance
 - EA Prerequisite: Fundamental Refrigerant Management
 - EA Credit: Optimize Energy Performance
 - EA Credits: Advanced Energy Metering
 - EA Credit: Renewable Energy
 - EA Credit: Enhanced Refrigerant Management
 - MR Prerequisite: Storage and Collection of Recyclables
 - EQ Prerequisite: Minimum Indoor Air Quality Performance
 - EQ Prerequisite: Environmental Tobacco Smoke Control
 - EQ Credit: Enhanced Indoor Air Quality Strategies
 - EQ Credit: Low-Emitting Materials
 - EQ Credit: Construction Indoor Air Quality Management Plan
 - EQ Credit: Indoor Air Quality Assessment
 - EQ Credit: Thermal Comfort
 - EQ Credit: Interior Lighting
 - EQ Credit: Daylight
 - EQ Credit: Quality Views
 - EQ Credit: Acoustic Performance

Provide the guidelines to all tenants before signing the lease.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M EA credit Energy Performance and WE credit Water Performance: Core and shell projects that require tenants to demonstrate a specified percentage reduction in energy and

water use can include the assumed tenant reduction in the core and shell energy and water models. Such requirements must be incorporated into the tenant lease and can be accurately measured by installing whole building energy and water meters. It is a simple yet critical strategy for understanding total energy and water use of tenants throughout the building life cycle. Tracking energy use and water consumption on a regular basis supports effective building management and provides data to help verify that building systems are operating as designed.

SS Credit: Places of Respite

This credit applies to

- ▶ BD+C: Healthcare (1-4 points)

INTENT

To provide patients, all staff, and visitors with the health benefits of the natural environment by creating outdoor places of respite on the healthcare campus.

REQUIREMENTS

HEALTHCARE

Provide places of respite that are accessible to patients and visitors, equal to 5% of the *net usable program area* of the building.

Provide additional dedicated places of respite for staff, equal to 2% of the net usable program area of the building.

Places of respite must be outdoors, or be located in interior atria, greenhouses, solariums, or conditioned spaces; such interior spaces may be used to meet up to 30% of the required area if 90% of each qualifying space's gross floor area achieves a direct line of sight to unobstructed views of nature.

All areas must meet the following requirements.

- ▶ The area is accessible from within the building or located within 200 feet (60 meters) of a building entrance or access point.
- ▶ The area is located where no medical intervention or direct medical care is delivered.
- ▶ Options for shade or indirect sun are provided, with at least one seating space per 200 square feet (18.5 square meters) of each respite area, with one wheelchair space per five seating spaces.
- ▶ Horticulture therapy and other specific clinical or special-use gardens unavailable to all building occupants may account for no more than 50% of the required area.
- ▶ Universal-access natural trails that are available to visitors, staff, or patients may account for no more than 30% of the required area, provided the trailhead is within 200 feet (60 meters) of a building entrance.

Additionally, outdoor areas must meet the following requirements.

- ▶ A minimum of 25% of the total outdoor area must be planted with two or more adapted or native vegetation types, or have *overhead vegetated canopy*. Monocultures, such as conventional grass lawns or turfgrass, do not count towards this requirement.
- ▶ The area is open to fresh air, the sky, and the natural elements.
- ▶ Signage must meet the 2010 FGI Guidelines for Design and Construction of Health Care Facilities (Section 1.2-6.3 and Appendix A1.2-6.3:Wayfinding).
- ▶ Places of respite may not be within 25 feet (7.6 meters) of a smoking area (see EQ Prerequisite Environmental Tobacco Smoke Control).

Existing places of respite on the hospital campus may qualify if they otherwise meet the credit requirements.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Definitions

- ▶ **monoculture** the cultivation of a single plant species in a given area.

Connection to Ongoing Performance

- ▶ LEED O+M SS Credit Rainwater Management: Rain gardens and other rainwater management features may also serve as outdoor places of respite.

SS Credit: Direct Exterior Access

This credit applies to

- ▶ BD+C: Healthcare (1-4 points)

INTENT

To provide patients, all staff, and visitors with the health benefits of the natural environment by creating outdoor places of respite on the healthcare campus.

REQUIREMENTS

HEALTHCARE

Provide direct access to an exterior courtyard, terrace, garden, or balcony. The space must be at least 5 square feet (0.5 square meters) per patient for 75% of all inpatients and 75% of qualifying outpatients whose clinical length of stay (LOS) exceeds four hours.

Patients whose length of stay exceeds four hours, and whose treatment makes them unable to move, such as emergency, stage 1 surgical recovery, and critical care patients, may be excluded.

Places of respite outside the building envelope that meet the requirements of SS Credit Places of Respite that are immediately adjacent to clinical areas or with direct access from inpatient units may be included.

Qualifying spaces must be designated as nonsmoking. The spaces must also be located more than 100 feet (30 meters) from building exhaust air locations, loading docks, and roadways with idling vehicles.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M SS Credit Rainwater Management: Rain gardens and other rainwater management features may also serve as outdoor spaces that meet the requirements of this credit.

SS Credit: Joint Use of Facilities

This credit applies to

- ▶ BD+C: Schools (1-4 points)

INTENT

To integrate the school with the community by sharing the building and its playing fields for nonschool events and functions.

REQUIREMENTS

SCHOOLS

Option 1. Make Building Space Open to the General Public (1 point)

In collaboration with the school authorities, ensure that at least three of the following types of spaces in the school are accessible to and available for shared use by the general public:

- ▶ auditorium;
- ▶ gymnasium;
- ▶ cafeteria;
- ▶ one or more classrooms;
- ▶ playing fields and stadiums; and
- ▶ joint parking.

Provide access to toilets in joint-use areas after normal school hours.

OR

Option 2. Contract with Specific Organizations to Share Building Space (1 point)

In collaboration with the school authorities, contract with community or other organizations to provide at least two types of dedicated-use spaces in the building, such as the following:

- ▶ commercial office;
- ▶ health clinic;
- ▶ community service centers (provided by state or local offices);
- ▶ police office;
- ▶ library or media center;
- ▶ parking lot; and
- ▶ one or more commercial businesses.

Provide access to toilets in joint-use areas after normal school hours.

OR

Option 3. Use Shared Space Owned by Other Organizations (1 point)

In collaboration with the school authorities, ensure that at least two of the following six types of spaces that are owned by other organizations or agencies are accessible to students:

- ▶ auditorium;
- ▶ gymnasium;
- ▶ cafeteria;
- ▶ one or more classrooms;
- ▶ swimming pool; and
- ▶ playing fields and stadiums.

Provide direct pedestrian access to these spaces from the school. In addition, provide signed joint-use agreements with the other organizations or agencies that stipulate how these spaces will be shared.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance:

- ▶ LEED O+M LT credit Alternative Transportation: Shared school parking lots or parking lots on neighboring sites used for school functions may reduce the number of new parking spaces, which may help improve a project's transportation performance score